AWS DevOps Syllabus

S. No.	Parameter	Detail
1	Course Title	AWS DevOps
2	Course Code	BP044
3	Duration (Hours)	75
4	No. of Theory Hours	25
5	No. of Practical Hours	50

Course Description:

The AWS DevOps Course delivers 75 hours of hands-on learning in automating, deploying, and managing applications on AWS. Covering CI/CD pipelines, Infrastructure as Code, containerization, and monitoring, learners gain expertise using tools like CodePipeline, CloudFormation, Docker, and Kubernetes. The program prepares participants for the AWS Certified DevOps Engineer – Professional exam and real-world cloud automation roles.

- **DevOps Fundamentals & AWS Overview**

  - ✓ DevOps culture, principles, and lifecycle
  - ✓ AWS DevOps tools overview
  - ✓ CI/CD concepts and release pipelines
  - ✓ Set up AWS account and IAM roles
  - ✓ Install and configure AWS CLI
  - ✓ Explore AWS Developer Tools suite

- **Source Code Management with AWS CodeCommit & Git**

  - ✓ Version control systems and Git essentials
  - ✓ CodeCommit architecture and access models
  - ✓ Create and configure a CodeCommit repository

ELEVATE ACADEMY

Copyrights by Elevate Academy

- ✓ Connect Git with CodeCommit and push code
- ✓ Manage branches and pull requests

➢ **Build Automation with AWS CodeBuild**

- ✓ Continuous Integration fundamentals
- ✓ Build environments, runtime, and buildspec.yml
- ✓ Create CodeBuild project and configure IAM policies
- ✓ Automate builds for a sample application
- ✓ Integrate testing into CodeBuild pipeline

➢ **Continuous Deployment with CodeDeploy**

- ✓ Application deployment strategies (Blue-Green, Canary)
- ✓ Deployment groups and AppSpec files
- ✓ Set up EC2/On-premise deployments via CodeDeploy
- ✓ Automate app updates and rollbacks
- ✓ Implement Blue-Green deployment on EC2

➢ **CI/CD Orchestration using CodePipeline**

- ✓ Continuous Delivery and release workflow
- ✓ CodePipeline integration with AWS DevOps services
- ✓ Create an end-to-end pipeline using CodeCommit, CodeBuild, and CodeDeploy
- ✓ Add manual approvals and notifications
- ✓ Trigger automatic deployments from Git commits

➢ **Infrastructure as Code (IaC) with CloudFormation & Terraform**

- ✓ IaC fundamentals and benefits
- ✓ CloudFormation templates and stacks
- ✓ Introduction to Terraform on AWS
- ✓ Build reusable CloudFormation templates
- ✓ Automate VPC, EC2, and S3 provisioning
- ✓ Write Terraform scripts for multi-environment deployment

➢ **Configuration Management with AWS OpsWorks & Ansible**

ELEVATE ACADEMY

Copyrights by Elevate Academy

- ✓ Configuration automation concepts
- ✓ OpsWorks layers, stacks, and Chef recipes
- ✓ Deploy web app using OpsWorks Stack
- ✓ Automate configuration tasks with Ansible playbooks
- ✓ Manage versioned deployments

➢ **Containerization & Orchestration (ECS, EKS, Docker)**

- ✓ Docker fundamentals and container lifecycle
- ✓ Amazon ECS and EKS architecture
- ✓ CI/CD for containerized apps
- ✓ Build and push Docker images to Amazon ECR
- ✓ Deploy containers using ECS Fargate
- ✓ Automate container deployment via CodePipeline

➢ **Monitoring, Logging & Performance Optimization**

- ✓ Observability in DevOps
- ✓ AWS CloudWatch, CloudTrail, and X-Ray overview
- ✓ Create CloudWatch dashboards and alarms
- ✓ Analyze application logs and metrics
- ✓ Trace application performance using AWS X-Ray

➢ **Security, Compliance & IAM Best Practices**

- ✓ AWS Identity & Access Management for DevOps
- ✓ Secrets management and encryption best practices
- ✓ Configure IAM roles for CI/CD pipelines
- ✓ Manage credentials with AWS Secrets Manager
- ✓ Implement least privilege access model

➢ **Automation & Scripting with AWS CLI and SDK (Python/Boto3)**

- ✓ AWS SDKs, Boto3, and automation principles
- ✓ Write Python scripts using Boto3 for resource automation
- ✓ Automate EC2 creation and monitoring
- ✓ Integrate Python scripts into CI/CD workflows

➢ **Cost Management & DevOps Optimization**

ELEVATE ACADEMY

Copyrights by Elevate Academy

- ✓ AWS pricing models and DevOps cost optimization
- ✓ Use AWS Budgets and Cost Explorer
- ✓ Optimize build and pipeline resource usage

➢ **Capstone Project – End-to-End DevOps Pipeline**

- ✓ Create a **CI/CD pipeline** integrating CodeCommit, CodeBuild, CodeDeploy, and CodePipeline
- ✓ Deploy containerized microservices to ECS or EKS
- ✓ Implement CloudWatch monitoring, IAM security, and cost optimization
- ✓ Present architecture documentation and results

Practical

DevOps Fundamentals & AWS Setup

1. Create an AWS account and set up IAM admin user
2. Install and configure AWS CLI on your local system
3. Explore AWS Developer Tools (CodeCommit, CodeBuild, CodeDeploy, CodePipeline)

Source Code Management with CodeCommit & Git

4. Create a Git repository and connect it with AWS CodeCommit
5. Clone, push, and manage branches using Git CLI
6. Configure repository access permissions and webhooks

Build Automation with CodeBuild

7. Create and configure a CodeBuild project for a Node.js app
8. Write and use a buildspec.yml file for automated testing
9. Integrate CodeBuild with CodeCommit for triggered builds
10. View build logs and analyze build failures

Continuous Deployment with CodeDeploy

ELEVATE ACADEMY

Copyrights by Elevate Academy

11. Create an application and deployment group in CodeDeploy
12. Deploy an EC2-based app using AppSpec.yml
13. Configure Blue-Green deployment for zero-downtime updates
14. Roll back to a previous version using CodeDeploy
15. Deploy a containerized app via CodeDeploy on ECS

CI/CD Orchestration using CodePipeline

16. Create an end-to-end pipeline using CodeCommit, CodeBuild, and CodeDeploy
17. Add approval stages and SNS notifications in CodePipeline
18. Automate pipeline triggers on every Git push
19. Integrate CodePipeline with GitHub repositories
20. Deploy a full-stack app with automated build and test stages

Infrastructure as Code (IaC) with CloudFormation & Terraform

21. Create a simple CloudFormation template for EC2 and S3
22. Automate VPC creation using CloudFormation stacks
23. Launch multi-tier architecture (EC2 + RDS + ELB) using templates
24. Write and execute a Terraform script to create AWS resources
25. Implement version-controlled IaC templates in a pipeline

Configuration Management with OpsWorks & Ansible

26. Create a stack and layer in OpsWorks for a web application
27. Deploy Chef recipes for system configuration automation
28. Write and run an Ansible playbook on EC2 instances
29. Integrate Ansible automation into CodePipeline

Containerization & Orchestration (ECS, EKS, Docker)

30. Install Docker and build a container image for an app
31. Push the Docker image to Amazon Elastic Container Registry (ECR)
32. Deploy containers using Amazon ECS with Fargate
33. Create a Kubernetes cluster using EKS
34. Deploy and scale microservices on EKS
35. Automate ECS deployments via CodePipeline

ELEVATE ACADEMY

Copyrights by Elevate Academy

Monitoring, Logging & Observability

36. Configure CloudWatch metrics and dashboards for EC2 and RDS
37. Set up CloudWatch alarms and SNS notifications
38. Enable and analyze logs with CloudTrail
39. Implement AWS X-Ray for distributed tracing
40. Automate performance reports for deployed applications

Security, Compliance & IAM Best Practices

41. Create IAM roles and policies for CodePipeline and CodeBuild
42. Secure secrets and credentials using AWS Secrets Manager
43. Apply encryption using AWS KMS
44. Enable GuardDuty and Security Hub for threat detection

Automation & Scripting with AWS CLI & SDK (Python/Boto3)

45. Automate EC2 creation and termination using Python (Boto3)
46. Write a script to create S3 buckets and upload files
47. Generate CloudWatch alarms using automation scripts
48. Integrate Boto3 scripts into CI/CD pipelines

Cost Management & Optimization

49. Configure AWS Budgets and receive alerts via SNS
50. Analyze resource usage and optimize costs using Cost Explorer

ELEVATE ACADEMY

Copyrights by Elevate Academy